

Technical Resolution Report

Project: VSC Manga App

Date: June 3, 2026

1. What was the Error?

The primary issue was that manga could not be saved to the library. In the MySQL database, the `saved_manga` table had a strict **Foreign Key Constraint** (`saved_manga_ibfk_2`) on the `manga_id` field.

This constraint required that any `manga_id` being saved must already exist in a local `manga` table. Since the application uses dynamic IDs from the MangaDex API and the local `manga` table was empty, the database rejected every save attempt with a `1452 Cannot add or update a child row` error.

2. How was it Resolved?

I performed three key actions to fix the application:

- **Database Fix:** I executed a SQL command to drop the failing constraint:

```
ALTER TABLE saved_manga DROP FOREIGN KEY saved_manga_ibfk_2;
```

- **Sign-In Logic Fix:** Updated `server.js` to allow users to log in using either their email OR their username.

```
db.query("Select * from users where email = ? OR username = ?", [email, username])
```

- **UI Display Fix:** Corrected the user object returned during login so the Navbar correctly displays the user's name (mapped from `user.username`).

3. Why was this Necessary?

This resolution was necessary for several reasons:

1. **Functional Integrity:** Without removing the database constraint, the core "Save to Library" feature was completely broken because it relied on a local data synchronization that doesn't exist in the current architecture.
2. **User Experience:** Allowing username login is a standard expectation for modern apps. Previously, users who tried to log in with their username would get a "User not found" error, which was confusing.
3. **Personalization:** Fixing the name mapping ensures that the app feels "alive" and confirms to the user that they are successfully logged in by greeting them in the Navbar.

Generated by Gemini CLI Agent